Umberto Cappellazzo

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Work Experience

NatWest Group London, UK
GenAI Research Engineer July 2026–Present

- Managers: Maja Pantic, Stavros Petridis

Imperial College London London, UK
Research Associate in the iBUG group (leader: Maja Pantic) March 2025–July 2026

- I have mainly worked on advancing multimodal speech recognition (i.e., auditory/visual/audio-visual speech recognition) through the integration of Large Language Models (LLMs), with an emphasis on parameter-efficient, adaptive, and architectural design strategies. Most of these works stem from collaborations with Meta AI. I also worked on large-scale self-supervised audio pre-training via next-embedding autoregressive objectives (i.e., obtaining strong audio learners via causal pretraining in the embedding space) and elastic variable-frame-rate speech tokenizers.
- Key projects developed over the past two years:
  * **Llama-AVSR** [*IEEE ICASSP 2025*]
  * **Llama-SMoP** [*INTERSPEECH 2025*]
  * **Llama-Matryoshka** [*IEEE ASRU 2025*]
  * **MoME: Mixture of Matryoshka Experts** [*NeurIPS 2025*]
  * **Omni-AVSR** [*IEEE ICASSP 2026*]
  * **Attention Sinks and Massive Activations in AVSR** [*IEEE ICASSP 2026*]
  * **Dr. SHAP-AV** [*INTERSPEECH 2026*]
  * **VIB-AVSR** [*INTERSPEECH 2026*]
- Supervisor: Stavros Petridis (ICL/NatWest Group)

Imperial College London London, UK
Research Intern, *Audio-visual speech recognition meets LLMs* February 2024–November 2024
Supervisor: Stavros Petridis (ICL/Meta AI)

- I investigated the efficient integration of LLMs for the task of audio-visual speech recognition. This culminated in `Llama-AVSR`, a multimodal LLM with strong audio-visual speech recognition abilities. This work has been accepted at *ICASSP* 2025. More details here.

Jelinek Summer Workshop on Speech and Language Technology (JSALT) Le Mans, France
Junior researcher in the FST group June 2023–August 2023

- Junior researcher for the "*Finite state methods with modern neural Architectures for speech applications and beyond*" group at JSALT2023 in Le Mans, France. I worked on the integration of early-exit techniques to make the training and inference of CTC/MMI systems dynamical. This led to a publication at ICASSPW 2023.

Education

University of Trento Trento, Italy
Ph.D. in Information Engineering and Computer Science [with honours] Nov. 2021–Jan. 2025

- Title: *Efficient Knowledge Transfer and Adaptation for Speech and Beyond*
- Advisors: Daniele Falavigna, Alessio Brutti

– Research interests: continual learning for audio and speech processing; multi-modal (i.e., audio-language) continual learning; parameter-efficient transfer learning of audio/speech (e.g., Adapters, Mixture of Adapters, LoRA); multi-modal LLMs for audio-visual speech recognition. This led to 9 publications in top-notch conferences.

University of Padua Padua, Italy
MSc in Telecommunication Engineering [110/110 with honours] 2016–2019

– Advisors: Michele Rossi, Matteo Gadaleta
– Thesis Title: A Deep Learning-Based ECG Delineator: Evaluation and Comparison on Standard Databases

University of Padua Padua, Italy
BSc in Information Engineering 2013–2016

– Advisor: Nicola Laurenti
– Thesis Title: Message Authentication over an Ideal or Noisy Channel

Technical Skills

- **Speech & Audio:** Auditory speech recognition (ASR), audio-visual speech recognition (AVSR), audio/speech/audio-visual foundations models (Audio Spectrogram Transformers, Whisper/WavLM/HuBERT, AV-HuBERT), spoken language understanding, self-supervised audio/speech representation learning.
- **(Multimodal) Large Language Models:** Llama-based multimodal LLMs, parameter-efficient supervised fine-tuning (LoRA, adapters, soft mixture of adapters), preference-based post-training (DPO, GRPO).
- **ML Architectures/Paradigms:** Mixture of Experts (MoE), knowledge distillation, matryoshka representation learning, contrastive learning, continual learning, variational information bottleneck.
- **Large-Scale Training:** PyTorch, Pytorch Lightning, distributed training (DDP, FSDP), large-scale data processing.

Talks & Presentations

- "Parameter-Efficient Fine-tuning for Audio and Speech Processing." *Invited talk at the CUED Speech Group Seminars at the University of Cambridge* (April 2024).

Professional Services & Mentorship

Reviewer

- Conferences: NeurIPS, ICLR, ACMMM, ICASSP, Interspeech, IJCNN, IEEE MLSP, CVPR.
- Journals: IEEE Signal Processing Letters, Transactions on Audio, Speech and Language Processing, Neurocomputing, International Journal of Computer Vision, Transactions on Image Processing, Knowledge-based Systems.

Mentorship

- Navlika Singh and Piyush Arora [Imperial College London, MSc students][winter 2025/2026]: *Variational information bottleneck in LLM-based AVSR.*
- Salman Hussain Ali [Mila, Quebec-AI-Institute, MSc student][2025]: *Lightweight mamba-based adapters for parameter-efficient transfer learning in speech and audio.*
- Anand [University of British Columbia, MSc student][summer 2025]: *Attention sinks and massive activation in audio-visual LLMs.*
- Lidia Prokopovych [MIT, B.A. student][summer 2025]: *Robustness evaluation of Llama-AVSR under different acoustic and visual noise.*
- Stefano Ciapponi [University of Bologna, MSc student][summer 2023]: *Prompting techniques to parameter-efficiently fine-tune neural models for speech processing.*

Publications

[1] Anand, U. **Cappellazzo**, S. Petridis, and M. Pantic, "Mitigating Attention Sinks and Massive Activations in Audio-Visual Speech Recognition with LLMs", *IEEE ICASSP*, 2026.

[2] Anand, U. **Cappellazzo**, S. Petridis, and M. Pantic, "Modality contribution dynamics in audio-visual speech recognition under visual perturbations", *under review*, 2026.

[3] P. Arora, N. Singh, U. **Cappellazzo**, S. Petridis, and M. Pantic, "VIB-AVSR: Variational Information Bottleneck for Noise-Robust LLM-Based Audio-Visual Speech Recognition", *INTERSPEECH*, 2026.

[4] U. **Cappellazzo**, X. Liu, P. Ma, S. Petridis, and M. Pantic, "Omni-AVSR: Towards Unified Multimodal Speech Recognition with Large Language Models", *IEEE ICASSP*, 2026.

[5] U. **Cappellazzo**, X. Liu, S. Petridis, and M. Pantic, "Listening Forward: Next Patch Embedding Prediction Enables Scalable Audio Learners", *arxiv preprint*, 2026.

[6] U. **Cappellazzo**, S. Petridis, and M. Pantic, "Dr. SHAP-AV: Decoding Relative Modality Contributions via Shapley Attribution in Audio-Visual Speech Recognition", *INTERSPEECH [Long Track]*, 2026.

[7] S. Hussain Ali, U. **Cappellazzo**, and M. Ravanelli, "MambAdapter: Lightweight Mamba-Based Adapters for Parameter-Efficient Transfer Learning in Speech and Audio", *INTERSPEECH*, 2026.

[8] J. Park, U. **Cappellazzo**, Z. Lin, A. Haliassos, H. Ma, Z. Wang, J. F. Magnotti, M. S. Beauchamp, J. Kwon, N. Harte, S. Petridis, and M. Pantic, "Phoneme-level visual benefit in modern avsr: A human–model comparison", *under review*, 2026.

[9] N. Singh, P. Arora, U. **Cappellazzo**, L. Della Libera, S. Petridis, and M. Ravanelli, "Elastic speech coding via matryoshka representation learning", *under review*, 2026.

[10] U. **Cappellazzo**, M. Kim, H. Chen, P. Ma, S. Petridis, D. Falavigna, A. Brutti, and M. Pantic, "Large Language Models are Strong Audio-Visual Speech Recognition Learners", *IEEE ICASSP*, 2025.

[11] U. **Cappellazzo**, M. Kim, P. Ma, H. Chen, X. Liu, S. Petridis, and M. Pantic, "MoME: Mixture of Matryoshka Experts for Audio-Visual Speech Recognition", *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[12] U. **Cappellazzo**, M. Kim, and S. Petridis, "Adaptive Audio-Visual Speech Recognition via Matryoshka-Based Multimodal LLMs", *IEEE ASRU*, 2025.

[13] U. **Cappellazzo**, M. Kim, S. Petridis, D. Falavigna, and A. Brutti, "Scaling and Enhancing LLM-based AVSR: A Sparse Mixture of Projectors Approach", *INTERSPEECH*, 2025.

[14] U. **Cappellazzo**, D. Falavigna, and A. Brutti, "Efficient Fine-tuning of Audio Spectrogram Transformers via Soft Mixture of Adapters", *Interspeech*, 2024.

[15] U. **Cappellazzo**, D. Falavigna, A. Brutti, and M. Ravanelli, "Parameter-Efficient Transfer Learning of Audio Spectrogram Transformers", *IEEE MLSP Workshop*, 2024.

[16] U. **Cappellazzo**, E. Fini, M. Yang, D. Falavigna, A. Brutti, and B. Raj, "Continual Contrastive Spoken Language Understanding", *ACL Findings*, 2024.

[17] G. A. Wright, U. **Cappellazzo**, S. Zaiem, D. Raj, L. Ondel Yang, D. Falavigna, and A. Brutti, "Training dynamic models using early exits for automatic speech recognition on resource-constrained devices", *Self-supervision in Audio, Speech and Beyond (SASB) Workshop, ICASSP*, 2024.

[18] M. Yang, U. **Cappellazzo**, X. Li, S. Watanabe, and B. Raj, "Improving continual learning of acoustic scene classification via mutual information optimization", *IEEE ICASSP*, 2024.

[19] M. Yang, X. Li, U. **Cappellazzo**, S. Watanabe, and B. Raj, "Towards Unified Evaluation of Continual Learning in Spoken Language Understanding", *INTERSPEECH*, 2024.

[20] U. **Cappellazzo**, D. Falavigna, and A. Brutti, "An Investigation of the Combination of Rehearsal and Knowledge Distillation in Continual Learning for Spoken Language Understanding", *INTERSPEECH*, 2023.

[21] U. **Cappellazzo**, M. Yang, D. Falavigna, and A. Brutti, "Sequence-Level Knowledge Distillation for Class-Incremental End-to-End Spoken Language Understanding", *INTERSPEECH*, 2023.

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